

METADRON CAPITAL

Systematic Multi-Asset Investment Platform

Executive Summary

Metadron Capital is a fully automated quantitative investment platform that combines macro regime analysis, fundamental security selection, and machine learning intelligence to generate alpha across equities, options, and futures. The system continuously scans approximately 1,600 securities, processes signals through multiple independent analytical tracks, and executes via institutional-grade algorithmic order routing.

The platform is designed for autonomous operation during market hours with continuous learning feedback that improves signal accuracy and execution quality over time.

Investment Philosophy

- **Beta managed within a 7–12% return corridor** aligned with S&P 500 historical earnings yield
 - **Alpha extracted through top-down macro regime classification** flowing into bottom-up security selection
 - **Fundamentals-first approach**: 40% of the decision weight is allocated to fundamental quality (ROIC, FCF, Graham-Dodd valuation, credit quality)
 - **Risk-adjusted position sizing** via Kelly criterion with volatility normalisation
 - **Options allocated only on detected mispricing** — fair value computed via Black-Scholes and Monte Carlo simulation, requiring a minimum edge before any position is taken
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Universe Coverage

Asset Class	Scope	Purpose
US Equities	S&P 500 + S&P 400 + S&P 600 (~1,500)	Core alpha generation
UK Equities	FTSE 100 representatives	International diversification
Options	Selected securities (edge-gated)	Convexity + mispricing capture
Futures	ES, NQ, YM, RTY, VX, ZN, ZB	Beta corridor management
Fixed Income	G10 + India sovereign, US IG/HY corporate	Macro signal + hedging
Commodities	GLD, SLV, USO, UNG, DBA, DBC + 3 more	Cyclical pattern signals
Currencies	G10 + INR + JPY	Macro regime context
Econometrics	40+ FRED series (GDP, CPI, M2, SOFR)	Regime classification

Total coverage: ~1,600 securities scanned continuously during market hours.

Signal Generation Architecture

The platform operates two independent parallel analytical tracks that converge before any investment decision is made:

Track A — Macro-Driven Fundamental Analysis

A sequential pipeline that begins with Federal Reserve balance sheet analysis, flows through macro regime classification, and fans out into specialised signal engines:

- **Fed Liquidity Analysis:** SOFR, reserves, TGA, ON-RRP, money velocity tracking
- **Macro Regime Engine:** Global Monetary Tension Framework with 4 non-linear gamma multipliers, sector ranking, yield curve analysis, credit pulse monitoring
- **10-Layer Intelligence Tensor:** Combines liquidity state, risk state, and capital flow into a unified regime signal (TRENDING / RANGE / STRESS / CRASH)
- **Parallel Signal Engines:** Graham-Dodd-Klarman security analysis, cross-asset contagion modelling (21-node graph), statistical arbitrage (mean-reversion + cointegration), distressed asset screening (5-model ensemble), fixed income analysis, and pattern discovery (symbolic regression)

Track B — News-Driven Event Intelligence

An independent pipeline that processes real-time news from 10,000+ sources and runs agent-based market simulation on flagged securities:

- **Real-Time News Processing:** WebSocket feed with sentiment scoring and urgency categorisation
- **Agent-Based Market Simulation:** Kyle Lambda and Heterogeneous Agent Model simulations per news-flagged ticker
- **Combined Score:** 40% news sentiment + 60% agent simulation direction, feeding into event-driven and contingent value rights analysis

Both tracks run independently with zero cross-dependencies and converge at the intelligence layer.

Intelligence & Decision Layer

Machine Learning Ensemble

A 10-tier weighted voting ensemble where each tier contributes an independent directional vote:

Tier	Model	Focus
1	Neural Network	Price direction prediction
2	Momentum / Mean-Reversion	Technical momentum signals
3	Volatility Regime	Vol compression / expansion
4	Monte Carlo	Probabilistic risk assessment
5	Fundamental Quality	Graham-Dodd investment grading
6	News + Agent Simulation	Real-time sentiment + microstructure
7	Distressed Asset	Credit distress + recovery detection
8	Event-Driven	M&A arbitrage, PEAD, catalysts
9	Contingent Value Rights	5-model CVR valuation
10	Credit Quality	Credit scoring (AAA–D)

Alpha Scoring Pipeline

All signals from both tracks converge into a dual alpha scoring engine:

- **Standard Pipeline:** Gradient boosted models + linear regression + CAPM alpha extraction with 22 engineered features
- **Enhanced Pipeline:** Walk-forward validation + 50+ factor library + sector-level mean-variance optimisation

Output: a scored, deduplicated, allocation-ready slate of opportunities.

4-Gate Quality Filter

Every potential trade must pass a cross-asset quality filter before execution:

Gate	Weight	Function
Fundamentals	40%	Quality, ROIC, FCF, credit, earnings
Flow & Headlines	20%	ETF flow, news sentiment, sector rotation
Macro Regime	20%	Direction alignment, VaR headroom
Momentum	20%	RSI, MACD, breakout confirmation

Minimum composite score of 0.55 required. The Fundamentals gate must pass independently — no trade proceeds without fundamental conviction regardless of momentum or sentiment.

Portfolio Construction

Allocation Structure

Sleeve	Target	Description
IG Equities	40%	Investment grade, all market caps
HY Equities	10%	BB-B rated, leveraged opportunities
Distressed Equity	10%	Fallen angels, recovery plays
Cashflow ETFs	15%	Monthly distribution vehicles (DRIP)
Fixed Income / Macro	5%	FI signals + macro relative value
Event-Driven / CVR	10%	M&A arb, PEAD, contingent value
Options (notional)	25%	Edge-gated only (≥ 200 bps mispricing)
Futures (notional)	15%	Beta corridor hedging (ES/NQ/VX)
Margin	8%	Initial margin for derivatives overlay
Cash Reserve	2%	Dry powder (hard floor, never breached)

Options Discipline

Options are treated as a precision instrument, not a speculative tool:

1. **Black-Scholes** theoretical pricing establishes fair value
2. **Monte Carlo simulation** (10,000 paths) estimates win probability
3. **Fair value vs market price** must show ≥ 200 basis points of edge
4. **Kelly criterion** sizes the position based on detected mispricing
5. **Minimum 5 contracts** — no token positions
6. **No edge detected** → **no allocation** regardless of available budget

Risk Management

- **Kill Switch**: 20% portfolio drawdown triggers automatic halt (requires manual operator reset)
- **10-Gate Pre-Trade Risk Check**: Position size, sector concentration, daily loss, gross/net leverage, trade throttle, drawdown, cash sufficiency, options delta, futures notional — all must pass

- **Beta Corridor:** Portfolio beta managed within 7–12% return corridor with volatility normalisation and Kalman-filtered smoothing
- **Per-Position Drawdown:** 20% individual position loss triggers automatic liquidation

Execution

Algorithmic Order Routing

All orders route through a unified execution surface with native institutional-grade algorithms:

Order Size	Algorithm	Method
> \$50,000 notional	TWAP	Time-weighted server-side splitting
Standard	VWAP	Volume-weighted market participation
Medium urgency	Adaptive	Broker selects optimal strategy
High urgency	Market	Immediate fill

Execution Quality

- **Micro-price estimation** adjusts limit prices based on order flow imbalance before submission
- **Transaction cost analysis** decomposes every fill into spread, market impact, timing, and commission components
- **Execution learning loop** continuously optimises routing strategy, slice count, and timing per context bucket (ticker × product × signal × regime × time-of-day × volatility × order size)
- **Trade log** records every generated order for reconciliation — full audit trail of what the system intended vs what executed

Continuous Learning

The platform operates a closed-loop feedback system with 7 independent learning channels:

Channel	What It Learns	What It Adjusts
Signal Accuracy	Which signals were profitable	ML ensemble tier weights
Execution Quality	Slippage patterns	Routing strategy selection
Regime Feedback	Regime prediction accuracy	HMM transition priors
Alpha Decay	How quickly alpha erodes	Position holding period
Risk Calibration	Risk event frequency	Gate thresholds
Agent Performance	Individual agent accuracy	Agent promotion / demotion
Cross-Asset	Sector allocation effectiveness	Macro sector weights

Weight changes are damped at $\pm 5\%$ per cycle with oscillation detection to prevent the system from chasing noise.

Agent Hierarchy

Autonomous sector-specialist agents are ranked and promoted based on sustained performance:

Rank	Requirements	Autonomy
Director	Sharpe > 2.5, accuracy > 85%	Full autonomy
General	Sharpe > 2.0, accuracy > 80%	High autonomy
Captain	Sharpe > 1.5, accuracy > 55%	Standard
Lieutenant	Sharpe > 1.0, accuracy > 50%	Restricted sizing
Recruit	Below thresholds	Under review

Monitoring & Observability

- **17 real-time metrics** exported for dashboard monitoring (NAV, leverage, risk level, kill switch status, slippage, fill latency, order counts by algo type)
- **Anomaly detection** across 8 dimensions (z-score, volume, correlation, VIX, credit, breadth)
- **Hourly CSV snapshots** of portfolio state for audit trail
- **Reconciliation engine** compares generated orders vs broker fills to detect any execution drift

Technology

Component	Implementation
Core Engine	Python (pure-numpy fallbacks, no single-framework dependency)
ML Models	XGBoost, scikit-learn, HMM (hmmlearn), pure-numpy PPO
Data Source	OpenBB (34+ providers: FMP, FRED, SEC, CBOE, Polygon)
Execution	Interactive Brokers (TWS/Gateway via ib_insync)
Options Pricing	Black-Scholes + Monte Carlo (10K path simulation)
Monitoring	Prometheus + Grafana
Process Management	PM2 (24/7 continuous operation)
Deployment	Hetzner dedicated server + Cloudflare SSL

The system is designed with graceful degradation — every external dependency is wrapped in try/except. If any component fails, the platform continues operating with reduced capability rather than halting entirely.

Key Differentiators

1. **Fundamentals-first, not momentum-driven:** 40% gate weight on fundamental quality prevents the system from chasing price action without underlying value
2. **Edge-gated options:** Unlike systems that deploy options budget mechanically, positions are only taken when quantitative mispricing is detected and sized by Kelly criterion
3. **Independent parallel tracks:** News-driven and macro-driven analysis run independently, preventing one track's noise from contaminating the other
4. **Institutional execution:** Native TWAP/VWAP algorithms with micro-price estimation and continuous execution learning
5. **Closed-loop learning:** Every execution outcome feeds back into the signal weights, ensuring the system adapts to changing market conditions without manual recalibration

6. **Full auditability:** Every signal, decision, order, and fill is logged with timestamps. The reconciliation engine provides complete transparency into what the system intended vs what executed.

Metadron Capital — Systematic Alpha Through Intelligent Automation